Match score model:
Poisson goals, outcomes, and empirical rates

Abstract

We specify independent Poisson goals for two teams, record the joint distribution and match-outcome probabilities, then connect the Poisson means (λ_A, λ_B) to abstract attack/defense strengths and to simple plug-in formulas using per-team goals for/against and a league baseline.

1 Match-level model: Poisson goals

For one fixture, let X_A and X_B denote goals scored by teams A and B over a fixed period (e.g. regulation time). We treat these as random counts.

1.1 Rates

Each team can be described by an *attack* strength and a *defense* strength (from data, priors, or ratings). The *expected* goals in the match are driven by how each side's attack meets the opponent's defense:

- λ_A reflects **A's attack against B's defense** (how well A scores when facing B's defensive quality).
- λ_B reflects **B's attack against A's defense**.

Abstractly one chooses a map from those four ingredients into two nonnegative numbers, e.g.

$$\lambda_A = f(\mathrm{atk}_A, \mathrm{def}_B), \qquad \lambda_B = f(\mathrm{atk}_B, \mathrm{def}_A),$$

with the exact form of f (linear, log-linear, tabulated, etc.) left to the modeller.

1.2 Independence and joint distribution

Assume

$$X_A \sim \mathrm{Poisson}(\lambda_A), \qquad X_B \sim \mathrm{Poisson}(\lambda_B), \qquad X_A \perp X_B \ \text{ given } (\lambda_A, \lambda_B). \tag{1}$$

Then the **joint probability mass function** is, for nonnegative integers i, j,

$$\mathbb{P}(X_A = i,\, X_B = j) = e^{-(\lambda_A+\lambda_B)}\, \frac{\lambda_A^i}{i!}\, \frac{\lambda_B^j}{j!}. \tag{2}$$

Computation. For implementation, one often truncates to a finite goal grid (e.g. $i, j \in \{0, \ldots, K\}$) and renormalizes the restricted distribution if probabilities outside the grid are negligible.

1.3 Match outcomes

From (2),

$$\begin{aligned} \mathbb{P}(A \text{ wins}) &= \mathbb{P}(X_A > X_B), \\ \mathbb{P}(\text{draw}) &= \mathbb{P}(X_A = X_B), \\ \mathbb{P}(B \text{ wins}) &= \mathbb{P}(X_B > X_A). \end{aligned}$$

These three probabilities sum to 1.

Summary of Section 1. Pick (λ_A, λ_B) from **A's attack vs B's defense** and **B's attack vs A's defense**, then use independent Poisson goals to obtain the full score distribution and win/draw/lose probabilities.

2 Interpreting λ as expectations

For a fixture between A and B, one may interpret

$$\lambda_A = \mathbb{E}[X_A \mid \text{match } (A, B)], \qquad \lambda_B = \mathbb{E}[X_B \mid \text{match } (A, B)],$$

when the Poisson model is taken literally. The rest of the note discusses *constructing* numerical values for λ_A, λ_B from simple league summaries.

3 Notation for empirical summaries

From past results (same competition and time window):

- $\bar{s}_i$: mean goals *for* team i per match (GF).
- $\bar{c}_i$: mean goals *against* team i per match (GA).
- μ: league mean goals scored per team–match row. With a balanced schedule and full double counting of each fixture, this typically equals the league mean GA per row as well.

4 Multiplicative attack × opponent defense

A simple closed form that adapts λ to the opponent is

$$\lambda_A \;=\; \bar{s}_A \cdot \frac{\bar{c}_B}{\mu}, \qquad \lambda_B \;=\; \bar{s}_B \cdot \frac{\bar{c}_A}{\mu}. \tag{3}$$

Interpretation. $\bar{s}_A$ is how much A scores against an "average" defense in the sample. The factor $\bar{c}_B/\mu$ measures how much *more* (or less) than average B concedes. If B is average defensively, $\bar{c}_B = \mu$ and $\lambda_A = \bar{s}_A$.

Equivalent form.

$$\lambda_A \;=\; \mu \cdot \frac{\bar{s}_A}{\mu} \cdot \frac{\bar{c}_B}{\mu},$$

i.e. baseline μ times an attack index for A times a defensive leakiness index for B.

Link to Section 1. This choice of f uses *observed* GF/GA as proxies for attack and opponent-relative defense; it is one convenient plug-in, not the only map from $(\mathrm{atk}_A, \mathrm{def}_B)$ to λ_A.

5 Log-linear (Dixon–Coles style) view

A standard parameterization is

$$\log \lambda_{ij} \;=\; \log \mu + \alpha_i + \delta_j,$$

where α_i is an attack effect and δ_j a defense effect on goals opponents score against j. Plug-in estimates from summaries (before centering) give $\alpha_i \approx \log(\bar{s}_i/\mu)$, $\delta_j \approx \log(\bar{c}_j/\mu)$, hence

$$\lambda_A \approx \mu \, \exp(\alpha_A + \delta_B),$$

consistent with (3) when exponentials are expanded without centering constraints. In a full fitted model, $\{\alpha_i\}$ and $\{\delta_j\}$ are usually centered for identifiability.

6 Shrinkage

Raw $\bar{s}_i$ and $\bar{c}_i$ are noisy when the sample size n_i is small. An empirical-Bayes style fix shrinks toward the league mean, e.g.

$$\tilde{s}_i \;=\; \frac{n_i}{n_i + k}\, \bar{s}_i + \frac{k}{n_i + k}\, \mu$$

(and similarly $\tilde{c}_i$), then substitute $\tilde{s}$ and $\tilde{c}$ into (3). The constant $k > 0$ controls shrinkage strength.

7 Further remarks

- **Consistency with** (1)**.** Under independent Poissons, λ_A and λ_B are not forced to sum to a fixed number; richer dependence (e.g. Dixon–Coles adjustment for low scores) modifies the joint law beyond (2).
- **Home advantage.** If desired, multiply the relevant λ by a factor $h > 1$ for the home side (or add $\log h$ in log space).
- **Schedule strength.** Raw $\bar{s}_i, \bar{c}_i$ mix team ability with the quality of opponents already played; iterative or likelihood-based attack–defense models partially address this.
- **Relation to** (2)**.** Any (λ_A, λ_B)—whether from theory, ratings, or (3)—can be inserted into (2) to approximate the score grid and outcome probabilities in Section 1.